ICT2113 · Data Structures and Algorithms

Core Structures: Lists, Stacks, Queues, and Trees

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

A stack follows last-in, first-out order; a queue follows first-in, first-out order. A tree represents hierarchical relationships, and a binary search tree maintains an ordering invariant that supports search when its shape remains balanced.

2. Example and reasoning

Choose the structure from the operations the problem needs. A browser history naturally uses a stack of prior states. A print queue needs arrival order. A search index may use a tree or hash table, depending on ordered queries and expected lookup behavior.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “core structures: lists, stacks, queues, and trees” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.